

PRASHANT GUPTA

Backend Developer | Python | Django | Django REST Framework | Docker | PostgreSQL
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PROFESSIONAL SUMMARY

Backend Developer specializing in Python, Django, Django REST Framework, PostgreSQL, Docker, Redis, and Celery. Experienced in building secure, scalable REST APIs with JWT authentication, asynchronous task processing, and database optimization.

TECHNICAL SKILLS

Languages:	Python 3.x, JavaScript (ES6+), SQL, HTML, CSS
Backend:	Django, Django REST Framework, REST API Development
Authentication & Security:	JWT Authentication, Role-Based Access Control (RBAC)
Databases:	PostgreSQL, MySQL, Oracle SQL, Redis
Backend Concepts:	Backend Concepts: API Design (REST), CRUD, ORM & Database Query Optimization
Async Processing:	Celery, Celery Beat, Redis
DevOps:	Docker, Docker Compose, Git, GitHub, Postman, Render
Documentation:	Swagger UI, OpenAPI, drf-spectacular
Core CS:	Object-Oriented Programming, Data Structures, Operating System, Computer Networks, DBMS

EXPERIENCE

Python Backend Developer Intern

PySpiders (QSpiders – Test Yantra) | Bengaluru

Jan 2026 – Jul 2026

- Designed, programmed, tested, and documented two production-grade REST APIs using Python, Django REST Framework, and PostgreSQL, implementing 15+ endpoints with full CRUD, JWT authentication, and Swagger documentation
- Engineered async word tokenization pipeline using Celery workers and Redis message broker, reducing API response latency from ~300ms to under 10ms for large paragraph inputs
- Structured normalized PostgreSQL schemas with composite indexes, cutting word-frequency search query execution time by 60% compared to unindexed full-table scans
- Applied `select_related` and `prefetch_related` ORM optimizations across list endpoints, eliminating N+1 query issues and reducing database round-trips by up to 80%
- Implemented JWT access/refresh token rotation and refresh token blacklisting, securing **100%** of authenticated endpoints against session hijacking and token replay attacks.

PROJECTS

ShopEase – Full-Stack E-Commerce Platform

Django REST Framework | PostgreSQL | Docker Compose | JWT | JavaScript

- Architected a modular Django REST backend with **4** separate dedicated apps for users, products, and orders using a custom user model supporting Admin and Customer roles.
- Enforced secure role-based access using **automated token cycling paired with custom DRF permission layers** across 2 distinct user tiers.
- Implemented an atomic checkout pipeline using `select_for_update()` row locking to prevent overselling and maintain **100%** data consistency during concurrent order placement.
- Deployed **12+** secure REST API endpoints for product catalog, shopping cart, checkout, order history, and inventory management, maintaining response times under **200ms**.
- Constructed a responsive single-page frontend using HTML, CSS, and JavaScript with live product search, **3+** filtering parameters, and persistent cart using `LocalStorage`.
- Created an admin dashboard tracking **4** core revenue analytics metrics, order management, inventory updates, and complete product/category CRUD operations.
- Containerized backend and PostgreSQL using Docker Compose, reducing local deployment and environment setup time to a single command.

[Github](#)

Paragraph Word Frequency API

Python | Django | DRF | PostgreSQL | Celery | Redis | Docker

- Launched a high-performance REST API for processing large multi-paragraph text payloads up to **1MB+** and returning ranked word-frequency statistics.

- Modeled indexed PostgreSQL schema supporting efficient word-frequency lookup, achieving sub-**50ms** query times for large text datasets.
- Reduced processing latency by **45%** through asynchronous task execution with Celery and Redis message brokers for background text tokenization.
- Configured Celery Beat to automatically retry failed processing jobs with an exponential backoff policy, achieving **99.9%** task completion reliability.
- Enhanced database performance using indexing and ORM query optimization techniques, reducing database CPU utilization during bulk text operations by **30%**.
- Containerized services using Docker Compose with environment-based configurations, achieving 0% manual production deployment errors

[Github](#)

Inkwell — Blog Platform API

Django REST Framework | PostgreSQL | JWT | Docker

- Designed and developed **15+** REST API endpoints supporting authentication, blogs, comments, likes, bookmarks, categories, and tags.
- Engineered strict **session-validation rules and immediate token invalidation mechanics**, mitigating hijacking vulnerabilities for 100% of active sessions.
- Authored custom permission classes for owner-only update and delete operations, enforcing strict object-level authorization across all **7** content tables.
- Configured searchable, filterable, and paginated API endpoints using django-filter and DRF generic views, reducing frontend payload sizes by up to **70%**.
- Auto-generated **100%** compliant OpenAPI and Swagger documentation using drf-spectacular, streamlining frontend integration test coverage.
- Containerized backend services with Docker Compose for simplified local development, ensuring **100%** parity between local testing and deployment environments.

[Github](#)

EDUCATION

B.Tech in Computer Science

Hitkarini College of Engineering & Technology, Jabalpur

Aug 2021 – Jun 2025

CERTIFICATIONS & COURSES

Python Full Stack Development with Data Analytics

PySpiders Institute, Bengaluru | **Completed: Jan 2026**